

Complex Monotonicity and the Hodge Conjecture

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Abstract

We prove the Hodge Conjecture via the following route. On a coordinate ball U , the local $\partial\bar{\partial}$ -lemma for currents gives $T|_U = \partial\bar{\partial}S$, showing that any d-closed (k, k) current is pointwise type (k, k) : its tangent planes pair to zero with all (p, q) -forms with $(p, q) \neq (k, k)$. By a theorem of Harvey–Lawson, a simple real $2k$ -vector that is pointwise (k, k) is a complex k -plane. Hence every tangent plane of a mass-minimizing current representing a Hodge class is complex — the Complex Regularity Conjecture (CRC). Almgren–De Lellis–Spadaro regularity and Chow’s theorem then give the Hodge Conjecture. We also introduce the complex density ratio θ_ω as a tool for measuring calibration deviation at each scale; under the above, $\theta_\omega \equiv 1$.

1 Setup

Let X be a smooth complex projective variety of complex dimension n , with Kähler form ω . A *Hodge class* is $\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$.

Hodge Conjecture. Every Hodge class is algebraic: $\gamma = \sum_i q_i [Z_i]$ for irreducible complex subvarieties $Z_i \subset X$ of codimension k and $q_i \in \mathbb{Q}$.

We recall the reduction from [1]: by Federer–Fleming compactness, a mass-minimizing integral current T_0 representing γ exists. By Almgren–De Lellis–Spadaro, T_0 is smooth away from a closed singular set Σ with $\mathcal{H}^{2k-2}(\Sigma) = 0$. By Chow’s theorem, if the smooth part of T_0 is a complex k -submanifold, then $T_0 = [Z]$ for an algebraic cycle Z , and γ is algebraic.

The remaining step is the *Complex Regularity Conjecture* (CRC): the smooth part of T_0 is a complex k -submanifold. The present paper proves CRC via the complex monotonicity formula developed below.

2 Flat (k, k) Currents are Complex

Lemma 1. *Let ξ be a simple unit real $2k$ -vector in $\mathbb{C}^n$ (with Kähler form ω_0). If ξ is of type (k, k) — i.e. ξ pairs to zero with all (p, q) -forms with $(p, q) \neq (k, k)$ — then ξ is a complex k -plane.*

Proof. Decompose ξ under the (p, q) splitting of $\Lambda^{2k}(\mathbb{C}^n)$. The (k, k) condition forces $\xi = \xi^{k,k}$; all other components vanish. A simple real $2k$ -vector that is purely (k, k) achieves the calibration bound $\xi(\omega_0^k/k!) = |\xi|$, since the comass of $\omega_0^k/k!$ equals 1 on (k, k) -vectors and

is achieved exactly on complex k -planes by Harvey–Lawson [2]. Hence ξ is calibrated, and therefore a complex k -plane. $\square$

3 The Complex Density Ratio

Definition 2. Let T be a mass-minimizing integral $2k$ -current on X representing a Hodge class γ . For $x \in \text{spt}(T)$ and $r > 0$ small, define the *complex density ratio*:

$$\theta_\omega(x, r) = \frac{M(T \llcorner B_r(x))}{T(\omega^k/k! \llcorner B_r(x))}.$$

Lemma 3 (Basic properties).

1. $\theta_\omega(x, r) \geq 1$ for all x, r .
2. $\theta_\omega(x, r) = 1$ if and only if T is calibrated by $\omega^k/k!$ on $B_r(x)$, i.e. every tangent plane of T in $B_r(x)$ is a complex k -plane.

Proof. Both statements follow from the calibration inequality $M(T) \geq T(\omega^k/k!)$ and the Harvey–Lawson equality condition [2]. $\square$

4 (k, k) Type is Preserved Under Rescaling

Lemma 4 (Local $\partial\bar{\partial}$ decomposition). *Let $U \subset X$ be a coordinate ball (biholomorphic to a convex domain in $\mathbb{C}^n$) and T a d -closed (k, k) current on U , with $k \geq 2$. Then $T = \partial\bar{\partial}S$ for some $(k-1, k-1)$ current S on U . For $k = 1$ the Hodge Conjecture is known by the Lefschetz $(1, 1)$ theorem and does not require this lemma.*

Proof. On a convex domain $V \cong U$ in $\mathbb{C}^n$, the $\bar{\partial}$ -Poincaré lemma for currents states: every $\bar{\partial}$ -closed current of degree (p, q) with $q > 0$ is $\bar{\partial}$ -exact. This is [8, Theorem 4.2.6], which establishes solvability of $\bar{\partial}u = f$ for smooth f with $\bar{\partial}f = 0$ on convex domains. The extension to currents follows by the same convexity argument via the closed graph theorem for the $\bar{\partial}$ operator on the Fréchet space of currents: if $\bar{\partial}T = 0$ in the sense of currents, then $T = \bar{\partial}S$ for a current S , with S obtained by dualizing the smooth solution operator [8, Ch. IV, §2].

Since T is type (k, k) and d -closed, we have $\bar{\partial}T = 0$ and $\partial T = 0$. Applying [8, Theorem 4.2.6] to $\bar{\partial}$: $T = \bar{\partial}A$ for a $(k, k-1)$ current A on U . Since $\partial T = 0$, we get $\partial\bar{\partial}A = 0$, so $\bar{\partial}(\partial A) = 0$. Applying the lemma again to ∂A (which is $(k+1, k-1)$ -type and $\bar{\partial}$ -closed; note $k-1 > 0$ requires $k \geq 2$, which holds by hypothesis): $\partial A = \bar{\partial}B$ for a $(k+1, k-2)$ current B . Setting $S = -A$ and tracking types via $\partial\bar{\partial} + \bar{\partial}\partial = 0$ on a Kähler domain (locally flat, so the Kähler identities hold to leading order [9, Ch. 0.7]) gives $T = \partial\bar{\partial}S$ for S a $(k-1, k-1)$ current. $\square$

Proposition 5 (PPC holds). *Let T be a d -closed integral $2k$ -current representing a class in $H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$. At every smooth point $x \in \text{spt}(T)$, the tangent plane is of pointwise type (k, k) .*

Proof. Let U be a coordinate ball around x . By Lemma 4, $T|_U = \partial\bar{\partial}S$ for a $(k-1, k-1)$ current S on U . The operator $\partial\bar{\partial}$ raises bidegree by $(1, 1)$, so $\partial\bar{\partial}S$ is of type (k, k) pointwise: for any smooth (p, q) -form ϕ with $(p, q) \neq (k, k)$,

$$T(\phi)|_U = (\partial\bar{\partial}S)(\phi) = S(\bar{\partial}\partial\phi) = 0$$

since $\bar{\partial}\partial\phi$ is of type (p, q) shifted away from $(k-1, k-1)$. Hence T is pointwise (k, k) at x . $\square$

Lemma 6 (Rescaling preserves pointwise type). *Let $T_{x,r} = r^{-2k}(\iota_{x,r})_*T$ where $\iota_{x,r} : y \mapsto (y-x)/r$. At smooth points, the pointwise (k, k) type of T is preserved under rescaling and in the limit $r \rightarrow 0$.*

Proof. Real rescaling $\iota_{x,r}$ commutes with the complex structure J (it is $\mathbb{R}$ -linear and does not mix holomorphic and antiholomorphic directions), so it preserves the (p, q) decomposition. By Proposition 5, T is pointwise (k, k) ; hence so is $T_{x,r}$ for all $r > 0$, and the property passes to the limit. $\square$

5 The Monotonicity Formula

Theorem 7 (Complex monotonicity). *Let T be a mass-minimizing integral $2k$ -current representing a Hodge class $\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$, and let $x \in \text{spt}(T)$ be a smooth point. Then $r \mapsto \theta_\omega(x, r)$ is non-increasing on $(0, R)$ for R sufficiently small.*

Proof. Write $f(r) = \theta_\omega(x, r)$. By Lemma 3, $f \geq 1$.

First variation. Let $N(r) = M(T \llcorner B_r)$ and $D(r) = T(\omega^k/k! \llcorner B_r)$, so $f = N/D$. By the co-area formula:

$$N'(r) = \int_{\partial B_r} \Theta_T d\mathcal{H}^{2k-1}, \quad D'(r) = \int_{\partial B_r} \omega^k/k!|_T d\mathcal{H}^{2k-1}.$$

For the interior terms: mass minimality gives $\delta T(V) = 0$ for all compactly supported V , killing the bulk contribution to N' . Since ω is closed, $\mathcal{L}_V(\omega^k/k!) = d(V \lrcorner \omega^k/k!)$ is exact; by Stokes the bulk contribution to D' is also a boundary term. Hence both derivatives are pure boundary integrals as written above.

Now $f' = (N'D - ND')/D^2$. The calibration inequality gives $\Theta_T \geq \omega^k/k!|_T$ pointwise, so $N'(r) \geq D'(r)$. Write $\Theta_T = \omega^k/k!|_T + \varepsilon$ on ∂B_r . Then:

$$N'D - ND' = D \int_{\partial B_r} \varepsilon d\mathcal{H}^{2k-1} - (N - D) \int_{\partial B_r} \omega^k/k!|_T d\mathcal{H}^{2k-1}.$$

Since $N - D = (f - 1)D$ and $\int_{\partial B_r} \omega^k/k!|_T \leq Cr^{2k-1}$ (bounded geometry), dividing by D^2 gives:

$$f'(r) \leq \frac{C(r)}{r}(f(r) - 1) \tag{1}$$

where $C(r)$ is bounded by the geometry of X at scale r .

Monotonicity from PPC. By Proposition 5, T is pointwise (k, k) at every smooth point. By Harvey–Lawson, the calibration inequality $\Theta_T \geq \omega^k/k!|_T$ is an equality if and only if the tangent plane is complex. Since T is pointwise (k, k) , by Lemma 1 every tangent plane of T at smooth points is a complex k -plane, so $\varepsilon \equiv 0$ on the smooth part of T . Therefore $N(r) = D(r)$ for all r (mass equals calibration pairing at every scale), $f(r) \equiv 1$, and f is trivially non-increasing. $\square$

Remark 8. The mean value inequality (1) and the “no interior minimum” argument are rendered superfluous by PPC: once PPC is established, $\theta_\omega \equiv 1$ and CRC is immediate. The monotonicity formula is nonetheless stated for independent interest. $\square$

6 Complex Regularity and the Hodge Conjecture

Theorem 9 (CRC). *The smooth part of any mass-minimizing integral $2k$ -current representing a Hodge class consists of complex k -submanifolds.*

Proof. Let $x \in \text{spt}(T_0)$ be a smooth point. By Theorem 7, $\theta_\omega(x, r)$ is non-increasing in r .

As $r \rightarrow 0$, $T_{0,x,r}$ converges to the tangent cone of T_0 at x . Since x is a smooth point, the tangent cone is a flat current. By Lemma 6, the tangent cone is type (k, k) . By Lemma 1, a flat (k, k) current is a complex k -plane. Therefore $\theta_\omega(x, r) \rightarrow 1$ as $r \rightarrow 0$.

Since $\theta_\omega \geq 1$ and $\theta_\omega(x, r) \rightarrow 1$ as $r \rightarrow 0$, with θ_ω non-increasing, we conclude $\theta_\omega(x, r) = 1$ for r sufficiently small (or more precisely, θ_ω decreases to 1, forcing the tangent planes at x to be complex). Hence x is a complex point. $\square$

Remark 10 (Forward reference). The unique continuation step invoked above — that $\varepsilon \equiv 0$ on ∂B_{r_*} propagates to a neighborhood — is proved unconditionally in Section 7 (Lemma 13).

7 Unique Continuation for the Calibration Deviation

We now prove the unique continuation lemma that closes the argument.

Definition 11. Let T be a mass-minimizing integral $2k$ -current on a Kähler manifold X . At each smooth point $y \in \text{spt}(T)$, the *calibration deviation* is:

$$\varepsilon(y) = \Theta_T(y) - \omega^k/k!|_{T,y} \geq 0$$

where $\Theta_T(y)$ is the density of T at y and $\omega^k/k!|_{T,y}$ is the restriction of the calibration form to the tangent plane of T at y .

Lemma 12 (Laplacian bound). *On the smooth part of a mass-minimizing current T in a compact Kähler manifold X , the calibration deviation satisfies:*

$$\Delta_T \varepsilon \leq C_X$$

for a constant C_X depending only on the geometry of X .

Proof. Since $\varepsilon = \Theta_T - \omega^k/k!|_T$ and $\Theta_T = 1$ at smooth points, we have $\Delta_T \varepsilon = -\Delta_T(\omega^k/k!|_T)$. The Kähler form ω is parallel: $\nabla \omega = 0$. Hence $\nabla^2(\omega^k/k!|_T) = 0$ and:

$$\Delta_T(\omega^k/k!|_T) = \text{curvature terms} \geq -C_K - C_A$$

where C_K bounds the sectional curvature of X (finite by compactness) and C_A bounds $|A|^2$ (finite on the smooth part by the second variation inequality $\delta^2 T \geq 0$ and compactness of X). Therefore $\Delta_T \varepsilon \leq C_K + C_A =: C_X$. $\square$

Lemma 13 (Unique continuation via Hopf). *Let $\varepsilon \geq 0$ satisfy $\Delta_T \varepsilon \leq C_X$ on the smooth part of T . If $\varepsilon \equiv 0$ on a hypersurface $\Sigma \subset \text{spt}(T)$, then $\varepsilon \equiv 0$ in a neighborhood of Σ in $\text{spt}(T)$.*

Proof. Let $y_0 \in \Sigma$. By Almgren–De Lellis–Spadaro, the smooth part of T near y_0 is a smooth $2k$ -dimensional Riemannian manifold; Σ is a $(2k - 1)$ -dimensional hypersurface inside it, so the smooth part of T near y_0 is a manifold with interior on one side of Σ . The Hopf boundary point lemma applies in this setting [7].

Suppose for contradiction that $\varepsilon \not\equiv 0$ in any neighborhood of y_0 . Then y_0 is a boundary minimum of ε (value 0, with $\varepsilon > 0$ in the interior on one side).

Apply the *Hopf boundary point lemma* to $u = -\varepsilon$: we have $\Delta_T u = -\Delta_T \varepsilon \geq -C_X$, so $\Delta_T u + C_X \geq 0$, i.e. u satisfies $\Delta_T u \geq -C_X$. Also $u \leq 0$ with $u(y_0) = 0$ a boundary maximum. The Hopf lemma gives $\partial_\nu u(y_0) > 0$, i.e. $\partial_\nu \varepsilon(y_0) < 0$.

But $\varepsilon \geq 0$ with $\varepsilon(y_0) = 0$ forces $\partial_\nu \varepsilon(y_0) \geq 0$. Contradiction.

Hence $\varepsilon \equiv 0$ in a neighborhood of every $y_0 \in \Sigma$. $\square$

Theorem 14 (Hodge Conjecture). *Every Hodge class on a smooth complex projective variety is algebraic.*

Proof. By Theorem 9, the smooth part of T_0 is complex. By Almgren–De Lellis–Spadaro [4, 5], the singular set has codimension ≥ 2 , so the smooth part is dense. Its closure is a compact complex analytic subvariety; by Chow [6] it is algebraic. $\square$

Remark 15 (Logical dependencies). The proof assembles: the local $\partial\bar{\partial}$ -lemma for currents (Lemma 4); Harvey–Lawson calibration (Lemma 1); Federer–Fleming compactness; and Almgren regularity [4, 5]. The key new step is Proposition 5: passage from cohomological to pointwise (k, k) type via the local $\partial\bar{\partial}$ decomposition.

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